3.1.3 Substances are exchanged between organisms and their environment by passive or active transport across exchange surfaces. The structure of plasma membranes enables control of the passage of substances across exchange surfaces.	
Cells	The structure of an epithelial cell from the small intestine as seen with an optical microscope. The appearance, ultrastructure and function of • plasma membrane, including cell-surface membrane • microvilli • nucleus • mitochondria • lysosomes • ribosomes • endoplasmic reticulum • Golgi apparatus. **Candidates should be able to** apply their knowledge of these features in explaining adaptations of other eukaryotic cells. The principles and limitations of transmission and scanning electron microscopes. The difference between magnification and resolution. Principles of cell fractionation and ultracentrifugation as used to separate cell components.
Plasma membranes	Glycerol and fatty acids combine by condensation to produce triglycerides. The R-group of a fatty acid may be saturated or unsaturated. In phospholipids, one of the fatty acids of a triglyceride is substituted by a phosphate group. The emulsion test for lipids. The arrangement of phospholipids, proteins and carbohydrates in the fluid-mosaic model of membrane structure. The role of the microvilli in increasing the surface area of cell-surface membranes.
Diffusion	Diffusion is the passive movement of substances down a concentration gradient. Surface area, difference in concentration and the thickness of the exchange surface affect the rate of diffusion. The role of carrier proteins and protein channels in facilitated diffusion. **Candidates should be able to** use the fluid mosaic model to explain appropriate properties of plasma membranes.
Osmosis	Osmosis is a special case of diffusion in which water moves from a solution of higher water potential to a solution of lower water potential through a partially permeable membrane. Candidates will **not** be expected to recall the terms hypotonic and hypertonic. Recall of isotonic will be expected.
Active transport	The role of carrier proteins and the transfer of energy in the transport of substances against a concentration gradient.
Absorption	Absorption of the products of carbohydrate digestion. The roles of diffusion, active transport and co-transport involving sodium ions.

Cholera	The cholera bacterium as an example of a prokaryotic organism. The structure of prokaryotic cells to include cell wall, cell-surface membrane, capsule, circular DNA, flagella and plasmid. Cholera bacteria produce toxins which increase secretion of chloride ions into the lumen of the intestine. This results in severe diarrhoea. The use of oral rehydration solutions (ORS) in the treatment of diarrhoeal diseases. **Candidates should be able to discuss** • the applications and implications of science in developing improved oral rehydration solutions • ethical issues associated with trialling improved oral rehydration solutions on humans.